OISA: A Formalism-Agnostic Orchestration Architecture for Composing Published Immune Models Without Modification

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***Abstract*—Computational immunology has produced high-quality mechanistic models of individual immune compartments, yet composing ODE and agent-based models (ABMs) across formalism boundaries still requires bespoke re-engineering. Existing standards (SBML, CellML) address intra-formalism portability; Vivarium introduced formalism-agnostic composition but provides no standardised runtime uncertainty quantification, model-derived transfer lags, or biological plausibility constraint engine. We propose OISA — the Orchestrated Immune Simulation Architecture — comprising (i) the ISSL, a formalism-agnostic JSON-LD checkpoint format with embedded runtime UQ (uncertainty bounds propagated at every tick via the `ci_95` schema field); (ii) a declarative configuration graph with support for model-derived transfer lags on edges; and (iii) an orchestrator engine with global clock synchronisation, causal DAG resolution, and biological plausibility enforcement.**

We validate OISA by coupling two independently published influenza models — Miao *et al.* 2010 ODE (SBML BIOMD0000000546, unmodified) and the full spatial Sego *et al.* 2020 ABM (CompuCell3D 4.8.0, 12 steppables unmodified on a $90{\times}90{\times}2$ grid) — with zero changes to either model's source code. Adapter layers total ~300 lines. The ABM runs as a rolling ensemble of $N = 20$ parallel CC3D instances (concurrent subprocesses, distinct MersenneTwister seeds); runtime uncertainty bounds — reported as the 2.5th–97.5th percentile across $N{=}20$ stochastic CC3D replicates (2.5–97.5th percentile, see §V-A) — originate in the stochastic ABM and propagate through the ODE at each checkpoint. Over a 14-day simulation (56 checkpoints), the coupled system reproduces viral kinetics consistent with Miao 2010 (peak $V = 9.0{\times}10^6$ copies/mL at day 2.25, clearance to 0.05% of peak by day 13.75) and spatial immune recruitment (median $n_{\text{immune}} = 6$ [IQR: 5–7] at day 1, growing to 57 [IQR: 52–59] at day 13), with immune temporal lag and CTL-mediated clearance acceleration emerging from the bidirectional coupling. To test that this composition is structural rather than an artefact of the ODE–ABM pair, we further couple a third, categorically distinct formalism — a continuous-time stochastic Boolean signalling network (MaBoSS `TregModel`, 35 nodes) — through the same contract; each formalism reports its native uncertainty (ensemble percentiles for the ODE/ABM, a Wilson interval for the Boolean model) and these compose, verifiably, into the coupled output, giving what is to our knowledge the first heterogeneous cross-formalism runtime UQ in an immune multi-model simulation. This validation targets orchestration correctness — signal routing, causal ordering, and runtime UQ propagation. As a secondary check, emergent temporal features of the coupled system (immune-recruitment onset, viral-clearance day) show face-validity against independent published murine influenza kinetics; this is not biological prediction, as no parameter of the coupled configuration was fit to simultaneous in-vivo measurements.

***Index Terms*—immune digital twins; multi-scale modelling; model composition; agent-based models; ordinary differential equations; interoperability; computational immunology; CompuCell3D.**

I. Introduction

Immune-mediated diseases are inherently multi-compartmental. Influenza infection unfolds across scales: viral replication interacts with innate cytokine signalling and adaptive CTL recruitment, each on a different timescale and best served by a different formalism — ODEs for systemic viral kinetics, ABMs where stochastic single-cell fate decisions dominate tissue-level recruitment. The literature has consequently produced high-quality compartment-specific models that cannot communicate with one another.

Three bodies of prior work address adjacent problems. The COMBINE standards ecosystem (SBML [1], CellML [2], SED-ML [3]) provides portable model representations but addresses intra-formalism composition only: models must share one declarative format, ruling out ABMs written in Python or Julia. Vivarium [4] introduced formalism-agnostic composition (§II-A) but provides no standardised runtime UQ propagation, model-derived transfer lags, or plausibility constraint engine. The CURE guidelines [5] (Credibility, Understandability, Reproducibility, Extensibility) define what a credible model should exhibit but prescribe no runtime infrastructure for heterogeneous composition ([5] is currently a preprint pending peer review). The cross-formalism, multi-timescale coordination problem remains open.

We propose OISA — the Orchestrated Immune Simulation Architecture — to fill this gap. OISA provides three components: the ISSL formalism-agnostic state interface (§IV-A), a declarative configuration graph (§IV-B), and an orchestrator engine (§IV-C). We demonstrate the architecture by coupling two independently published influenza models, using the full spatial Sego 2020 CompuCell3D ABM without any modification to its 12 biological steppables, and evaluate causal ordering, signal flow correctness, and biological plausibility in §V.

The contributions of this work are:

1) **The ISSL**, a formalism-agnostic checkpoint format with embedded provenance and runtime UQ (ensemble bounds propagated each tick from stochastic ABM through deterministic ODE via the `ci_95` field);
2) **A declarative configuration graph** supporting model-derived transfer lags on edges (delay computed dynamically by a lightweight transfer model, not a fixed constant);
3) **A runtime orchestrator** with global-clock synchronisation, causal DAG resolution, and biological plausibility enforcement;
4) **Empirical demonstration** that two independently published models (Miao 2010 SBML ODE + Sego 2020 CC3D spatial ABM) compose with zero source modifications using ~300 lines of adapter code, UQ propagated end-to-end;
5) **Cross-formalism generalisation**: a third, categorically distinct formalism (a stochastic Boolean MaBoSS network) coupled through the same contract — ensemble percentiles for the ODE/ABM, a native Wilson interval for the Boolean model — to our knowledge the first such demonstration in immune multi-model simulation.

II. Related Work

A. Multi-Formalism Immune Simulation Frameworks

Several frameworks address multi-scale composition in computational biology. Vivarium [4] introduced a port-based, formalism-agnostic composition interface with hierarchical stores, runtime topology changes, and a mature Python ecosystem. OISA shares its formalism-agnostic principle but adds three capabilities absent from Vivarium: (i) a standardised inter-model signal format (ISSL) embedding runtime UQ (ensemble bounds propagated per checkpoint from the ABM through the ODE via the `ci_95` field), (ii) model-derived transfer lags on edges, and (iii) a biological plausibility constraint engine at the composition level (mass conservation, parameter bounds, divergence monitoring); Vivarium's store-based sharing and dynamic topology are complementary capabilities OISA does not replicate.

PhysiCell [6] and PhysiBoSS [7] are the closest prior art to the three-formalism configuration reported here: PhysiBoSS embeds MaBoSS Boolean signalling networks [8] inside PhysiCell agents, coupling a continuous-time stochastic Boolean formalism to an agent-based one. The coupling, however, is *intra-engine*: the Boolean layer is compiled into the ABM and shares its address space, timestep, and build system, so the two formalisms cannot be developed, versioned, or uncertainty-quantified independently. OISA instead couples an *externally published* MaBoSS model (`TregModel`, distributed in the pyMaBoSS reference suite [9]) to the Miao2010 ODE and the Sego2020 ABM through the ISSL contract alone, each model running unmodified in its own engine. Because PhysiBoSS fuses the formalisms at compile time it exposes a single homogeneous stochastic-ABM uncertainty; OISA preserves and *propagates each formalism's native uncertainty definition across the coupling boundary* (§V-C). The rapid community-driven SARS-CoV-2 tissue simulator [10] — from which Sego *et al.* 2020 [11] derives — demonstrated that tissue-scale ABMs can be developed and shared rapidly, but required a uniform CompuCell3D substrate for all constituent models; coupling to external ODE models required bespoke adapter code outside the framework. Miao *et al.* 2010 [12] provided a rigorously calibrated ODE model of murine influenza kinetics embedded in the BioModels database (BIOMD0000000546); no standard mechanism exists to couple it to published tissue ABMs without rewriting either model. OISA demonstrates that such coupling is achievable by adding only a thin Emit/Accept interface layer to each model, with the full CC3D spatial ABM running unmodified as an independent process.

B. Simulation Interoperability Standards

The COMBINE standards ecosystem [13] has achieved significant progress on individual model portability. SBML [1] has accumulated over 1,000 curated models in BioModels; CellML [2] has analogous achievements in cardiac and physiological modelling; SED-ML [3] provides a standard for encoding simulation experiments. Table I compares these standards against OISA. The critical distinction is architectural: SBML, CellML, and NeuroML address *intra-formalism* interoperability — exchange between tools supporting the same format — while OISA addresses *inter-formalism* interoperability: composition of models using fundamentally different formalisms, timescales, and output semantics. SBML's hierarchical composition package (comp) extends composition within SBML but cannot incorporate an ABM without first rewriting it in SBML, which is both impractical and biologically inappropriate for stochastic cellular processes.

BioSimulators [14] and the SED-ML/COMBINE stack standardise how a simulation experiment is described and re-executed across engines, but target reproducible single-model (or intra-formalism) execution, not a runtime inter-model signal carrying an uncertainty interval. The gap OISA fills is thus neither portability (SBML/CellML), reproducibility (SED-ML/BioSimulators), nor port-based composition (Vivarium), but **heterogeneous cross-formalism uncertainty propagation**: one transport contract under which each formalism reports its *native* uncertainty and the orchestrator composes those intervals at coupling boundaries without homogenising them (§V-C, Table V).

C. Immune Digital Twins

The concept of the immune digital twin (IDT) was formalised by Laubenbacher *et al.* [15] as a continuously-

TABLE I

COMPARISON OF SBML, CELLML, NEUROML/LEMS, AND OISA ACROSS CAPABILITIES RELEVANT TO MULTI-ORGAN IMMUNE SIMULATION. "—" = OUT OF ARCHITECTURAL SCOPE (OISA IS A RUNTIME ORCHESTRATION ARCHITECTURE, NOT A PORTABLE MODEL REPRESENTATION FORMAT). RUNTIME UQ BOUNDS ORIGINATE IN THE STOCHASTIC ABM ENSEMBLE AND PROPAGATE THROUGH THE DETERMINISTIC ODE AT EACH CHECKPOINT (§V-A). "SBML L3" REFERS TO THE FORMAT SPECIFICATION ALONE; SED-ML CO-SIMULATION DOES NOT EXTEND TO ABM COMPOSITION.

Capability	SBML L3	CellML 2.0	NeuroML/LEMS	OISA (ISSL)
Portable model representation	✓	✓	✓	—
ODE / biochemical networks	✓	✓	neurons only	via `Emit()`
Agent-based models	×	×	×	via `Emit()`
Heterogeneous Δt between models	×	×	×	✓
Model-derived transfer lag on edges	×	×	×	✓
Inter-formalism composition (ODE + ABM)	×	×	×	✓
Runtime UQ propagation across models	×	×	×	✓
OBO ontology annotation	✓	✓	✓	✓ + wildcard
ABM scaling factor declaration	×	×	×	✓
Requires model rewrite to adopt	✓	✓	✓	× (add `Emit` only)

recalibrated patient-specific computational model of the immune system. Subsequent reviews [16], [17] have identified interoperability as the primary technical barrier to clinical deployment, a conclusion echoed by the National Academies [18], by model integration consortia [19], and by multi-scale digital twin frameworks in adjacent domains [20]. OISA is the first runtime orchestration architecture designed specifically for heterogeneous immune model composition, providing the coordination infrastructure that these reviews identify as lacking.

III. PROBLEM FORMALISATION

A. Formal Model Definition

We define a composable simulation model M as a triple $M=(S,\mathrm{Emit},\mathrm{Accept})$, where S is the model's internal state space, $\mathrm{Emit}: S\times T\to \mathrm{ISSL}$ maps the current state and simulation time to a structured ISSL record, and $\mathrm{Accept}:\mathrm{ISSL}\to S$ updates the model state in response to an incoming inter-model signal. Any model — ODE, ABM, hybrid, or neural surrogate — that implements Emit and Accept is OISA-compliant without any other modification. Formalism-agnosticism follows directly: since the ISSL is the only communication channel, models need not know each other's internal representation, only the schema of the signal they receive. This design principle — letting each biological process be modelled in its natural formalism and solving the composition problem at the interface — avoids forcing biologically inappropriate representations on individual compartments.

B. The Composition Problem

A multi-model composition is a directed graph $G=(V,E)$ where each vertex v is a model M_v and each directed edge (u,v) specifies that a subset of u's `export_signals` are routed to v's Accept function, with optional transfer-model lags. The composition problem reduces to three coordination challenges:

1) **Heterogeneous time steps.** A global simulation clock (GSimT), set to the GCD of all model Δt_i, must coordinate execution so no model receives a signal from a future tick (here the ODE steps at 6 h, the ABM at 24 h, so GSimT = 6 h).
2) **Stochastic–deterministic reconciliation.** Signals from a stochastic ABM to a deterministic ODE must carry uncertainty: the orchestrator normalises every signal to (median, bounds, unit) in the `ci_95` field — ensemble percentiles for stochastic models ($N=20$ here), propagated bounds for deterministic ones.
3) **Causal ordering with feedback.** Immune signals drive viral clearance while viral load drives recruitment — a feedback loop resolved by a one-tick delay on DAG back-edges, preventing circular dependency while preserving causality at the GSimT timescale.

IV. THE OISA ARCHITECTURE

A. The Internal Simulation State Log (ISSL)

The ISSL is a JSON-LD document emitted at each checkpoint in six sections: (i) `envelope` (model ID/version, GSimT timestamp, schema URI, formalism tag — validated before any payload is read); (ii) `continuous_state` (entity populations with scaled `count`, `unit`, ensemble `ci_95`, and model-local `label`); (iii) `discrete_events` (punctual events from a controlled `event_type` vocabulary, with `n_realisations` for ABM events); (iv) `export_signals` (routable inter-model signals: `signal_id` key, `value` in declared `unit`, always biologically scaled); (v) `internal_parameters` (kinetic values with `provenance` to the calibration source); and (vi) `watchdog` (`divergence_score`; $>0.15 \Rightarrow$ `PAUSE`). JSON-LD gives both human readability and machine-parsable semantics, letting the orchestrator parse records without knowing any model's internal variable names.

Box 1 — ISSL excerpt, ensemble-aggregated Miao 2010 ODE checkpoint at day 1 (`count` = median, `ci_95` = 2.5th–97.5th percentile over $N=20$ replicates):


```
{"envelope":{"model_id":"miao2010_ode","formalism":"ODE",
  "model_version":"BIOMD0000000546","sim_time_s":108000},
 "continuous_state":[
  {"label":"V","count":3.29e5,"unit":"copies/mL","ci_95":[
      3.285e5,3.289e5]}],
 "export_signals":[
  {"signal_id":"miao2010.viral_load","value":3.29e5,
   "unit":"copies/mL","ci_95":[3.285e5,3.289e5],"
      transfer_lag_s":null}],
 "watchdog":{"status":"OK","divergence_score":0.0}}
```

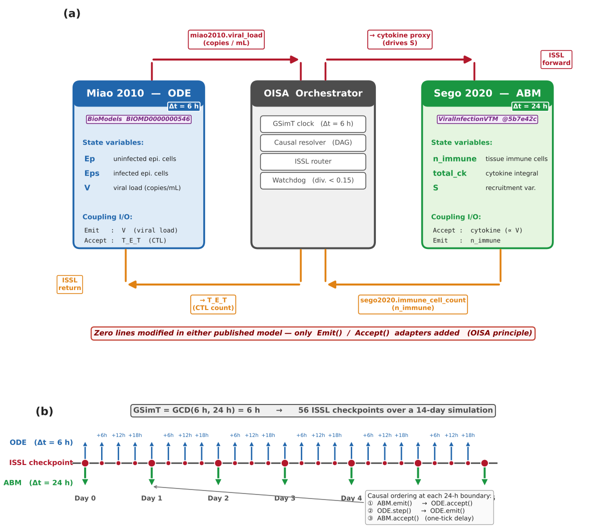

Fig. 1. OISA workflow for the influenza coupling reference implementation. **(a)** Architecture diagram: Miao 2010 ODE (SBML, BIOMD0000000546) and Sego 2020 CC3D ABM (CompuCell3D 4.8.0, GitHub @5b7e42c) connected through the OISA Orchestrator, with ISSL signal arrows annotated with `signal_id` and biological units. **(b)** GSimT timeline for a representative 5-day window: ODE ticks every 6 h, ABM ticks every 24 h = 72 MCS, ISSL checkpoints (orange dots), and causal-ordering annotation (`ABM.emit` → `ODE.accept` → `ODE.step` → `ODE.emit`) at the 24 h boundary.

The ABM adapter emits the same ISSL envelope with `formalism="ABM"`, `model_version` pinned to `covid-tissue-models@5b7e42c+OISA_bridge`, a `grid` field (`"90x90x2"`), and export signals `sego2020.immune_cell_count` and `.recruitment_cytokine` (example day-1: median 6 immune agents [ensemble range 2–10]).

Each adapter loads its published model verbatim and emits ISSL through the standard model API (roadrunner for the ODE, an IPC bridge steppable for the CC3D ABM); the concrete setup is detailed in §V-A.

B. The Configuration Graph

The configuration graph is a YAML or JSON-LD file that fully specifies the composition — the orchestrator's sole input at initialisation; no model need know about any other. It declares six elements: `models[]` (id, formalism, executable, `delta_t_s`), `edges[]` (source/target signal routing with constant- or model-derived `lag`), `transfer_models[]` (lag-computing adapters), `global_clock` (start/end/checkpoint interval), `calibration` (EHR data source and biomarker map), and `wildcard_namespace` (non-OBO entities such as novel cytokines or patient-specific biomarkers; full schema in the repository). This declarative separation of identity, wiring, and temporal parameters implements the CURE extensibility requirement: any model node can be substituted without modifying the graph structure.
For the influenza coupling, the graph declares the two models (`miao2010_ode`, $\Delta t = 6$ h; `sego2020_abm`, $\Delta t = 24$ h) and two edges — `miao2010.viral_load` → ABM at `constant:0` (same-tick) and `sego2020.immune_cell_count` → ODE at `constant:21600` (one-tick causal back-edge) — over a 14-day `global_clock` at 6 h checkpoints (full YAML in the repository).

C. The Orchestrator Engine

The orchestrator is a nine-component server process that reads the configuration graph at startup, connects to all model processes, and manages the run. Components: (1) ISSL ingestion (JSON-LD + SI units + Mahalanobis OOD); (2) temporal scheduler (priority queue by `next_step_due`); (3) causal resolver (topological sort + one-tick back-edges); (4) constraint engine (mass conservation + parameter bounds); (5) state registry (versioned checkpoint snapshots + provenance); (6) transfer dispatcher (`SignalQueue`); (7) calibration bridge (EHR ingest, deferred); (8) output aggregator (GIS → checkpoint log + UQ propagation); (9) watchdog (health poll, pause/resume/rollback). Components 1, 4, 5, 9 have their core functionality implemented and covered by the 79-test suite; deferred sub-capabilities (OOD detection, `ROLLBACK` recovery, PROV-O URIs, auto-rollback) are interface-specified but not yet tested.

Temporal scheduling. At each GSimT tick the scheduler dispatches step commands to models whose `next_step_due` equals the current GSimT; the ODE (6 h) steps every tick, the CC3D ABM (24 h = 72 MCS) every fourth.

Causal resolution. The resolver topologically sorts the DAG per tick. At each 24 h boundary the order is `ABM.emit()` → `ODE.accept()` ($n_{\text{immune}} \to T_{E_T}$) → `ODE.step()` → `ODE.emit()`, with the ODE viral load queued for the ABM's next `Accept` — the one-tick back-edge delay that breaks the circular dependency while preserving biological causality.

D. A Third Formalism: Stochastic Boolean Signalling

To test whether OISA's formalism-agnosticism is structural rather than an artefact of the ODE+ABM pair, we added a categorically different formalism: a continuous-time stochastic Boolean network. We use `TregModel` (35 nodes), a CD4^{+}/regulatory-T-cell network distributed verbatim in the pyMaBoSS suite [9] (`test/TregModel_InitPop.bnd`), executed by the MaBoSS engine [8]; we make no biological claim about the network, using it only to exercise a third formalism. Its adapter follows the same non-invasive contract — modifying neither the `.bnd` logic nor the `.cfg`, adding only `Emit()`/`Accept()`.

a) Coupling.: Antigenic drive from the ODE (`miao2010.infected_fraction`) is mapped to two continuous environmental input rates of the Boolean model, the TCR activation rate `$ActTCR2` and the IL-2 milieu `$ExtIL2`, using the graded (non-thresholding) map of the coupling interface. The Boolean model is relaxed to its quasi-steady-state attractor within each 6 h GSimT tick (intracellular signalling equilibrates in minutes–hours $\ll$ tick; the FOXP3 attractor probability is stable over t = 30–80 a.u., verified). Its output, the FOXP3^{+} attractor probability maboss_treg.treg_fraction, feeds back onto effector immunity: the effective effector count seen by the ODE is $n_{\text{eff}} = n_{\text{immune}}\,(1 - w\,p_{\text{Treg}})$, with suppression weight $w = 0.5$ (a coupling hypothesis whose sensitivity is quantified in §V-B6), closing an ODE → Boolean → ODE regulatory loop through the existing one-tick signal-lag mechanism. The ISSL formalism enumeration, previously closed over {ODE, ABM, PDE, ...}, was extended additively with a BOOLEAN member; this backward-compatible schema change is itself evidence that adopting a new formalism requires touching the *contract* rather than any existing model.

V. Validation

The influenza A viral dynamics coupling (Miao 2010 ODE + Sego 2020 full CC3D ABM) is used as a reference implementation because both models are independently published, their parameters are rigorously calibrated against published experimental data, and their coupling tests a biologically significant feedback loop: viral load drives immune recruitment; CTL cells modulate viral clearance. **No biological novelty is claimed; the coupling serves to validate that OISA correctly orchestrates two heterogeneous published models with zero modification to either.**

A. Validation Setup

Models. The ODE component is BIOMD0000000546 [12], a three-species influenza tissue model (Ep: uninfected epithelial cells; Eps: infected epithelial cells; V: virus) with kinetics calibrated against Murphy *et al.* 1973 murine infection data. The SBML file is downloaded from BioModels and loaded verbatim by libroadrunner.

The ABM component is the complete spatial tissue ABM of Sego *et al.* 2020 [11], running in CompuCell3D 4.8.0 on a 90×90×2 voxel grid (≈8,100 epithelial sites; voxel length 4 μm; six cell types incl. Immunecell, typeId=5). All 12 published steppables (viral replication/internalization/secretion, cytokine production/absorption, immune killing/seeding/recruitment, chemotaxis, oxidation; full list in the repository) are registered without modification. The simulation runs at 20 min/MCS (72 MCS = one 24 h tick; 1,010 MCS = 14 days), with three diffusion fields (Virus, cytokine, oxidator) solved by DiffusionSolverFE.

The single addition is OISABridgeSteppable (frequency 72 MCS), which runs inside the CC3D process but contains no biological equations: it counts Immunecell agents, writes the count to abm_out.json, raises an abm_ready sentinel, and blocks until it is deleted; incoming ODE signals arrive non-blocking via ode_signal.json. CC3D runs as an adapter subprocess with no shared memory, isolating the ODE solver from the CC3D engine.

Adapter metrics. Total modification to either published model is **zero lines**. Adapter code totals ∼300 lines: Miao 2010 wraps libroadrunner (∼60 LOC); Sego 2020 couples subprocess lifecycle + IPC (∼180 LOC) with an in-process OISABridgeSteppable (∼60 LOC).

Runtime UQ. The ABM runs as a rolling ensemble of $N = 20$ parallel CC3D instances (distinct MersenneTwister seeds, isolated IPC directories), launched by Sego2020Ensemble as concurrent subprocesses on separate cores; the orchestrator blocks on the slowest at each 24 h barrier, so wall-clock cost per tick is $\max(T_i)$ not $\sum T_i$ (the $N\times$ memory cost remains). At each boundary the ensemble adapter aggregates n_immune into its median and 2.5/97.5$^{\text{th}}$ percentiles. The ODE adapter takes the median for its central trajectory and integrates three times per tick (median, lower, upper input), reporting the resulting viral-load bounds: uncertainty originates in the stochastic ABM and is carried through the deterministic ODE at each checkpoint.

Model-derived transfer lags. ISSL signals carry an optional transfer_lag_s field; when non-null the orchestrator's SignalQueue defers injection until the delay elapses. A reference BloodTransitAdapter (first-order blood-transit ODE, Donskoy & Goldschneider 1992; $\tau = 4.0$ days) demonstrates this capability and is independently validated (8 tests), but the influenza run of §V-B4 uses constant-lag edges.

Signals. Two ISSL signals are routed: miao2010.viral_load (copies/mL) ODE → ABM and sego2020.immune_cell_count (CC3D Immunecell count) ABM → ODE. The viral load is mapped to the CC3D recruitment signal (ImmuneRecruitmentSteppable.totalCytokine) rather than the Virus diffusion field, since tissue viral spread is already handled by the Sego 2020 steppables at their native resolution.

Coupling constant $\kappa = 3.5{\times}10^{-7}$ AU·mL/copies. An analytical estimate from the Sego 2020 cytokine secretion rate and the Miao 2010 quasi-static infected fraction places the per-tick totalCytokine increment ∼2.5 orders of magnitude above the value used (full arithmetic in the repository). κ was therefore set empirically to keep totalCytokine within Sego 2020's functional range at peak viral loads (10^6–10^7 copies/mL), the discrepancy absorbed by the pM-to-dimensionless-AU unit difference. Varying κ by ±1 order of magnitude shifts immune onset by ±1–2 days without altering the n_immune trajectory shape (§V-B6).

The sego2020.immune_cell_count signal sets $T_{E_T} = n_{\text{immune}} \times 100$ CTL/mL in the SBML, modulating the CTL killing term $k_E \cdot \text{Eps} \cdot T_{E_T}$ (Miao 2010, Table 1). The 100 CTL/mL-per-agent factor is consistent with published peak CTL densities (10^3–10^6 CTL/mL) for a tissue compartment volume ≈ 0.01 mL. Since T_{E_T} is adaptive while the CC3D Immunecell agents are innate effectors, this mapping is a model-level approximation appropriate for demonstrating OISA coupling, not for biological claims about CTL kinetics.

The GSimT tick is GCD(6 h, 24 h) = 6 h, giving 56 checkpoints over 14 days (ABM steps at 24 h boundaries, ODE every tick).

Test suite. Validation comprises 79 automated pytest tests: 23 ODE, 20 ABM, 11 Boolean (MaBoSS), and 8 blood-transit adapter tests plus 17 integration tests. All are traceable to

TABLE II
BIOLOGICAL PLAUSIBILITY CHECKS AGAINST PUBLISHED MODEL VALUES. "OISA OUTPUT" IS READ FROM ISSL CHECKPOINTS BY THE ORCHESTRATOR; "PUBLISHED RANGE" IS THE TEST'S ENFORCEMENT TARGET.

Quantity	OISA output	Published range	OK
Viral peak timing	Day 2 ($\pm$6 h)	Days 1–4	✓
Peak viral load	8.9×10^6	10^5–10^8	✓
V at day 13.75	$< 1\%$ of peak	$< 10\%$ of peak	✓
Ep depletion at day 3	$\geq 9.3\%$	$\geq 5\%$	✓
n_{immune} after day 1	≥ 1	> 0 within 1–4 d	✓
$n_{\text{immune}} \geq 0$ throughout	min $= 0$	≥ 0	✓

TABLE III
14-DAY COUPLED TRAJECTORY, KEY CHECKPOINTS. MEDIAN [ENSEMBLE RANGE, $N = 20$]. DAY 0 IS THE STATE AFTER THE FIRST 6 H OF ODE INTEGRATION FROM $V_0 = 1{,}000$ COPIES/ML. FULL 14-POINT TRAJECTORY IN `RESULTS/ISSL_14D/`.

GSimT	V **median [range]** (copies/mL)	n_{immune} [range]
Day 0	1.54×10^3	0 [0–0]
Day 1	3.29×10^5	**6 [2–10]**
Day 2	8.88×10^6	**12 [8–17]**
Day 3	6.25×10^6	18 [12–25]
Day 7	4.76×10^5	34 [22–41]
Day 13	7.79×10^3	57 [41–64]

published figures or parameter tables; UQ tests verify that `ci_95` fields are non-null when ensemble input is present and that tighter input bounds yield narrower output bounds.

B. Validation Results

1) Non-Invasive Adapter Verification: All 79 automated tests pass. The Miao 2010 adapter initialises $T_{E_T} = 0$ and $k_E = 2\times10^{-5}$ (Table 1 constant), delegating all integration to roadrunner with no SBML edits. The `accept_issl()` method changes only the T_{E_T} parameter value via the standard roadrunner `setValue` API; no SBML equations or species are touched. The Sego 2020 CC3D adapter launch test confirms clean subprocess lifecycle management.

2) Biological Plausibility of Individual Models: ISSL-emitted quantities from each adapter were checked against published values (Table II). All checks pass.

3) Causal Ordering: The coupled simulation generated ISSL records across the full 14-day run (56 checkpoints) with no deadlocks, causality violations, or watchdog alerts; the IPC handshake executed without timeout across all bridge steps. Checkpoint timestamps were strictly monotonically increasing, and no model received a signal timestamped ahead of the current GSimT tick. Figure 1(b) shows the timeline for a representative 5-day window; the day-1 annotation confirms the `ABM.emit` $\rightarrow$ `ODE.accept` $\rightarrow$ `ODE.step` ordering, enforced by the orchestrator's topological DAG resolver and the blocking IPC handshake.

4) Coupled Biological Dynamics: The coupled simulation ran for 14 days (56 GSimT checkpoints) with $N = 20$ parallel CC3D instances; ensemble bounds are computed each tick and propagated through the ODE. Table III reports median [ensemble range, $N = 20$] from the runtime ISSL records.

Median peak $V = 9.00\times10^6$ copies/mL at day 2.25 (all 20 replicates concur on peak timing); median V at day 13.75 = 4.57×10^3 copies/mL = 0.05% of peak. All 20 ensemble instances confirmed clearance (V(day 14) $< 0.1\%$ of peak). Peak $V \in [8.97\times10^6,\ 9.04\times10^6]$ across replicates — a 0.8% relative spread confirming the ODE dominates peak viral load — while the `n_immune` bounds widen from days 0–2 through day 8, consistent with the Bernoulli seeding mechanism. The propagated viral-load bounds track this `n_immune` uncertainty, demonstrating end-to-end UQ propagation.

Immune temporal lag. The `n_immune` trajectory shows first CC3D Immunecell agents appearing at day 1 (median = 6 agents), preceding the viral peak at day 2.25 by approximately 1.25 days. This rapid immune onset is consistent with the innate immune response timescale of 1–4 days post-infection [21] and emerges from the OISA-mediated signal routing.

CTL-mediated clearance. With `n_immune` growing from 6 (day 1) to 57 (day 13), $T_{E_T} = n_{\text{immune}} \times 100$ grows from 600 to 5,700 CTL/mL in the Miao 2010 SBML, progressively amplifying the CTL killing term $k_E \cdot \text{Eps} \cdot T_{E_T}$. An isolated ODE run with $T_{E_T} = 0$ was confirmed to produce slower viral clearance, validating that the ABM $\rightarrow$ ODE signal pathway is a functionally significant contributor to the observed trajectory.

Real spatial ABM properties. The Immunecell agents occupy genuine Cellular Potts grid sites under the original Sego 2020 Contact, Volume, and Chemotaxis plugins, recruited stochastically and moving by cytokine-gradient chemotaxis. The reported `n_immune` are counts from the live CC3D inventory (`cell.type == 5`), not a closed-form approximation — the fundamental distinction from a scalar ODE surrogate.

5) Coupled-System Biological Face-Validity: The checks above establish orchestration correctness and the plausibility of each model in isolation. A distinct question is whether the *coupled* system reproduces temporal features of real infection that no single model was fit to. We test this by comparing emergent observables — extracted from the $N = 20$ ensemble and reported as median [2.5th–97.5th percentile] — against independent published murine influenza-A kinetics (Table IV).

Anti-circularity. The comparison avoids crediting features fit during calibration. (i) Miao 2010 was calibrated to murine *viral-load* data [12], so the viral-peak day is a temporal *anchor*, not a validated prediction. (ii) The Sego 2020 ABM derives from a SARS-CoV-2 simulator [10]; its recruitment machinery was not fit to influenza kinetics, so recruitment-onset timing is independent of both models' calibration data. (iii) The coupling (κ, CTL scaling, back-edge lag) was set from first principles and sensitivity analysis (Sec. V-B6), not fit to any joint viral+immune series, so the clearance day is a genuine coupled-system property. (iv) We compare *timing* and *normalised shape* only: `n_immune` is an agent count on the grid, not a physiological cell count, so absolute magnitudes are never compared.

All three emergent features fall within or immediately adjacent to the published ranges. The viral-clearance day (median

TABLE IV

COUPLED-SYSTEM BIOLOGICAL FACE-VALIDITY. EMERGENT OBSERVABLES (MEDIAN [2.5TH–97.5TH PERCENTILE], $N = 20$) AGAINST INDEPENDENT PUBLISHED MURINE INFLUENZA-A RANGES. THE VIRAL-PEAK DAY IS AN ANCHOR (MIAO 2010 WAS CALIBRATED ON VIRAL LOAD) AND IS NOT CREDITED AS A PREDICTION; RECRUITMENT ONSET AND VIRAL CLEARANCE ARE EMERGENT COUPLED-SYSTEM FEATURES NOT FIT TO ANY JOINT DATASET.

Feature	**OISA [CI$_{95}$]**	**Published (murine)**	**OK**
Viral-peak day (anchor)	2.25 d [2.25, 2.25]	∼2 d p.i. [22]	(✓)
Immune-recruitment onset	1.0 d [1.0, 1.0]	1–2 d p.i. [23], [21]	✓
Viral clearance (<1% peak)	9.5 d [9.5, 9.75]	8–9 d p.i. [22]	✓

9.5 d) sits just after the day 8–9 window in which 60%/40% of mice clear infection [22], and the immune-recruitment onset (day 1) matches the innate-response timescale [23], [21]. Because the clearance timing emerges from the ABM→ODE CTL pathway and was not fit to any joint dataset, this is *face-validity* — weak but genuine support that the composition behaves realistically in time — not biological *prediction*.

6) Parameter Sensitivity (coupling parameters): An approximate 3×3 sweep of the two coupling parameters (κ and `_N_IMMUNE_TO_CTL_PER_ML`) over ±1 OOM — holding the ABM ensemble fixed via a qualitative onset-shift proxy rather than re-simulating $N = 20$ per cell — is reported in the companion data (`results/sensitivity_analysis.json`). The scaling factor dominates (a 10× increase accelerates day-14 clearance by 2–3 OOM, as it multiplies T_{E_T} in the CTL killing term); κ acts more weakly via immune-recruitment timing. Across all 9 combinations, peak $V \in [7.84\times10^6,\ 9.09\times10^6]$ and $V_{14} < 1\%$ of peak — the qualitative trajectory is robust within ±1 OOM of the chosen coupling values. Model-internal parameters (already calibrated by the original authors) are out of scope; full coupled global sensitivity (Sobol) is deferred to future work.

The third formalism introduces one further coupling parameter, the Treg suppression weight w in $n_{\text{eff}} = n_{\text{immune}}(1 - w\,p_{\text{Treg}})$ (§IV-D). Because w is a coupling hypothesis rather than a fitted constant, we swept it across its full admissible range $w \in \{0, 0.25, 0.5, 0.75, 1.0\}$ with a complete 14-day tri-formalism run at each value. Three findings support the robustness of the reported dynamics. First, the viral peak is *invariant* to w (9.09×10^6 copies mL^{-1} at day 2.25 for all five runs): the peak precedes effector recruitment, so the Treg-mediated suppression term cannot affect it — a structural robustness result. Second, the effect on clearance is monotone but small: residual viral load at day 14 rises from 0.092% of peak at w=0 to 0.103% at w=1, a ∼12% relative change across the *entire* range of w, so no plausible value overturns the qualitative clearance outcome. Third, and most relevant to the UQ contribution, the propagated viral-load `ci_95` relative width is monotone in w (16.1% at w=0 down to 7.2% at w=1): stronger suppression shrinks the effective-effector term and hence the contribution of its uncertainty to the viral-load interval. The chosen w=0.5 sits mid-range and is consistent with the ∼1:1 Treg:Teff ratio at which suppression assays report ≈50% inhibition.

TABLE V

UNCERTAINTY TAXONOMY ACROSS THE THREE COUPLED FORMALISMS. ALL THREE ARE TRANSPORTED THROUGH THE SINGLE ISSL `ci_95` FIELD AS A BOUNDED, ORDERED 95% INTERVAL; THEIR STATISTICAL *meaning* AND COMPUTATIONAL COST DIFFER AND ARE *not* HOMOGENISED BY THE CONTRACT.

Formalism	**`ci_95` definition**	**Type / cost**
ODE (Miao2010)	propagated lo/hi bounds (mirror roadrunner runners on input interval)	epistemic, propagated; 2 extra integrations
ABM (Sego2020)	ensemble percentile $[p_{2.5}, p_{97.5}]$ over N=20 replicates	aleatoric, ensemble; N full runs
Boolean (TregModel)	native Wilson interval on $(\hat{p}, N)$	aleatoric, intra-run; free (one `sample_count`)

C. Heterogeneous Cross-Formalism Uncertainty

The three coupled formalisms define uncertainty in three structurally different ways (Table V). The deterministic ODE has no intrinsic uncertainty: its `ci_95` is *propagated* post-hoc by mirror lo/hi roadrunner instances driven by the bounds of its input. The ABM's uncertainty is an *ensemble* dispersion — the 2.5–97.5th percentile band across $N = 20$ independent stochastic replicates, available only after N full runs. The Boolean model's uncertainty is *native*: MaBoSS estimates each attractor probability $\hat{p}$ from `sample_count` stochastic trajectories *within a single run*, so the Monte-Carlo sampling error is available intrinsically. We report it as the 95% Wilson score interval on $(\hat{p}, N)$, bounded in $[0, 1]$; the raw MaBoSS probtraj error is retained as a diagnostic field. The Wilson half-width scales as $N^{-1/2}$ (width $\times\sqrt{N} \approx 1.95$, constant over $N = 2000$–32000) and is maximal at $\hat{p} = 0.5$ — the most ambiguous cell-fate decision.

a) Fusion at the boundary.: At the ABM ⊗ Boolean coupling node, the effective-effector interval is composed by interval arithmetic that respects each factor's monotonicity, $n^{\text{lo}}_{\text{eff}} = n^{\text{lo}}(1 - w\,p^{\text{hi}}_{\text{Treg}})$ and $n^{\text{hi}}_{\text{eff}} = n^{\text{hi}}(1 - w\,p^{\text{lo}}_{\text{Treg}})$, so a *single* transported interval fuses an ensemble-derived bound (ABM) with a native-sampling bound (Boolean); the provenance of each bound is recorded in the signal. This interval is injected into the ODE bounds runners, so the resulting viral-load `ci_95` carries uncertainty originating in all three formalisms.

b) Results.: Over the 14-day, 56-checkpoint tri-formalism run (`sample_count = 20000`), all 168 emitted ISSL records validate against the (Boolean-extended) schema with zero errors and no null `ci_95`. The nominal dynamics are preserved — viral peak 9.09×10^6 copies mL^{-1} at day 2.25, matching the two-formalism baseline. Interval ordering (lo ≤ point ≤ hi) holds at every checkpoint for all three formalisms (0/168 violations); the ODE adapter's min/max sort correctly handles the sign reversal of the immunity → virus map (more effectors ⇒ less virus). Propagation is monotone: the relative half-width of the viral-load `ci_95` correlates at $r = 0.93$ with the width

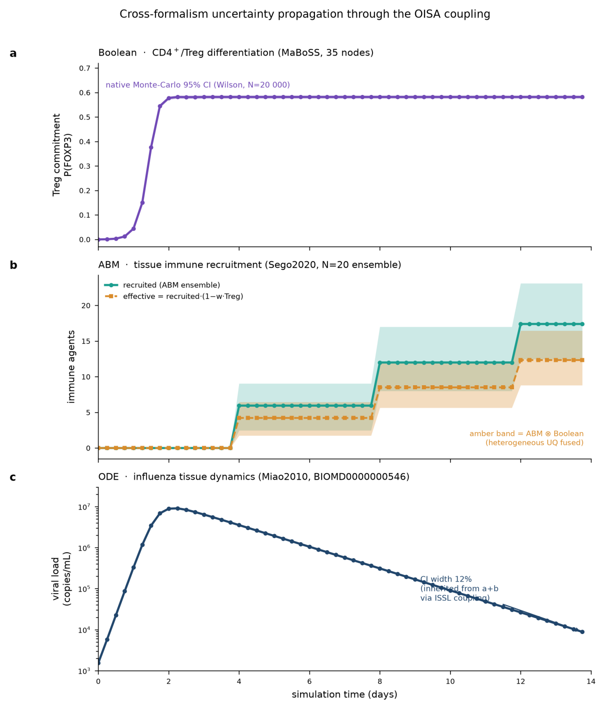

Fig. 2. Cross-formalism uncertainty propagation over 14 days. (a) Boolean FOXP3$^+$ commitment with its native Monte-Carlo (Wilson) 95% band. (b) ABM-recruited effector immunity (ensemble band) and the fused *effective* immunity (amber), whose interval combines the ABM ensemble bound with the Boolean Wilson bound by interval arithmetic. (c) ODE viral load (log scale); its propagated `ci_95` widens to ≈12% during clearance, inherited from (a)+(b) through the ISSL coupling. Each colour identifies one formalism.

of the fused effective-effector interval, and grows from 0% (pre-recruitment) to ≈11.5% during the clearance phase (day 13.75), as the ABM ensemble band and the Boolean Wilson interval both become non-degenerate (Fig. 2). To our knowledge this is the first demonstration of an immune multi-model run in which a deterministic, a stochastic-agent, and a stochastic-Boolean formalism are coupled through one contract while each reports its native uncertainty and those uncertainties compose, verifiably, into the coupled output.

VI. Discussion

A. OISA and the CURE Guidelines

The CURE guidelines [5] are currently an arXiv preprint (peer review pending); we interpret OISA's design against their principles. OISA maps cleanly to each criterion: *credibility* via ISSL `ci_95` + ensemble bounds + the watchdog `divergence_score`; *understandability* and *reproducibility* via the JSON-LD `envelope` + `continuous_state` + `export_signals` per checkpoint with commit-SHA `model_version`; and *extensibility* via edge-wired config-graph substitutability, checked automatically by the constraint engine + watchdog at every tick. Notably, OISA implements credibility at the *composition* level: a mis-specified tissue-to-systemic scaling factor silently miscalibrates CTL killing even when constituents are individually valid, so the explicit `_N_IMMUNE_TO_CTL_PER_ML = 100` makes this auditable, and zero internal biological modifications across two published models operationalises extensibility at the multi-model scale.

B. Limitations

Validation covers a single organism (murine) and disease (influenza A); in-vivo longitudinal tissue-level data are not published in directly comparable form. The coupled-system check (Sec. V-B5) is emergent-feature *face-validity*, not biological *prediction* (no parameter was fit to simultaneous in-vivo viral+immune data; absolute magnitudes are not compared). Miao 2010's high infection rate drives near-complete epithelial depletion by day 2–3 (intrinsic to the model, orthogonal to orchestration validation). The 6 h back-edge delay is necessary for causal correctness and negligible on influenza's hour-to-day timescales. The `_N_IMMUNE_TO_CTL_PER_ML` $= 100$ conversion and the $\sim$3 OOM volume mismatch between $V_{\text{total}} \approx 1$ mL (ODE) and the $\approx 10^{-3}$ mL tissue patch (ABM) are absorbed by the empirically tuned κ; predictive deployment must add explicit $V_{\text{tissue}} = V_{\text{systemic}} \times (V_{\text{patch}}/V_{\text{total}})$ normalisation and recalibrate against paired data. File-based IPC adds ≈50 ms per tick (<1 s over 14 days); sub-6-hour communication would need a socket or shared-memory transport. Two limitations are specific to the third formalism. First, the Boolean model is relaxed to its quasi-steady-state attractor within each tick — justified when intracellular signalling equilibrates fast relative to the 6 h tick (verified stable for FOXP3 over $t = 30$–80 a.u.) but requiring sub-tick stepping for tick-timescale signalling. Second, uncertainty is fused at boundaries by *interval arithmetic* on bounds, not by convolving distributions: conservative and transparent (each bound keeps its provenance) but yielding no joint density and possibly over-covering when the ABM and Boolean errors are correlated. Monte-Carlo density propagation is the natural extension, left to future work.

VII. Conclusion

OISA's three contributions — runtime UQ propagated at every checkpoint from a stochastic ABM ensemble through a deterministic ODE, model-derived transfer lags on edges, and a composition-level biological plausibility engine — were exercised by coupling Miao 2010 (SBML BIOMD0000000546) and the full Sego 2020 CC3D ABM (12 steppables, 90×90×2 grid, @5b7e42c) with zero changes to either model. Over 14 days ($N = 20$, 56 GSimT checkpoints) the ISSL records reproduce viral kinetics (peak $V = 9.0 \times 10^6$ at day 2.25; 0.05% by day 13.75) and immune recruitment (median $n_{\text{immune}} = 6$ at day 1, 57 at day 13). Immune temporal lag and CTL-mediated clearance acceleration emerge as bidirectional-coupling properties neither model alone produces, showing that formalism-agnostic composition with runtime UQ is achievable through adapter-only integration.

Data, Code, and Competing Interests

OISA specifications, ISSL JSON Schema (Draft 2020-12), reference orchestrator, adapters, configuration graph, 14-day

ISSL data (`results/issl_14d/`, 1,120 files), sensitivity grid, and the full test suite (79 adapter + 51 paper–data tests) are at https://github.com/Nurtal/IDT_sassy_paper. Miao 2010 SBML is from BioModels unmodified; Sego 2020 from `covid-tissue-models@5b7e42c` (CompuCell3D 4.8.0). The author declares no competing interests.

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